Machine basics, DC machines, rotating fields

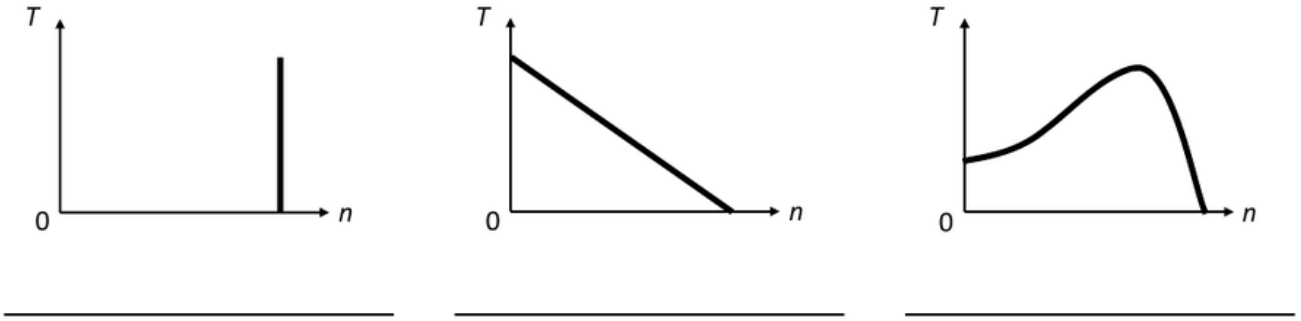
Practice questions · Day 07 · Fri 30 Oct · pack M6

5. Practice

From your course files

Jun24 Q3a (past exam, 6 marks). Torque vs. Speed Characteristics. Consider the torque versus speed curves shown in the figure below.

i. For each curve, write the type of machine in the space below the curve and draw the resulting torque curve if the no-load/synchronous speed was reduced to half of its original value. [3 marks]
ii. Alongside each curve, indicate how you could halve the no-load/synchronous speed while ensuring safe machine operation. [3 marks]



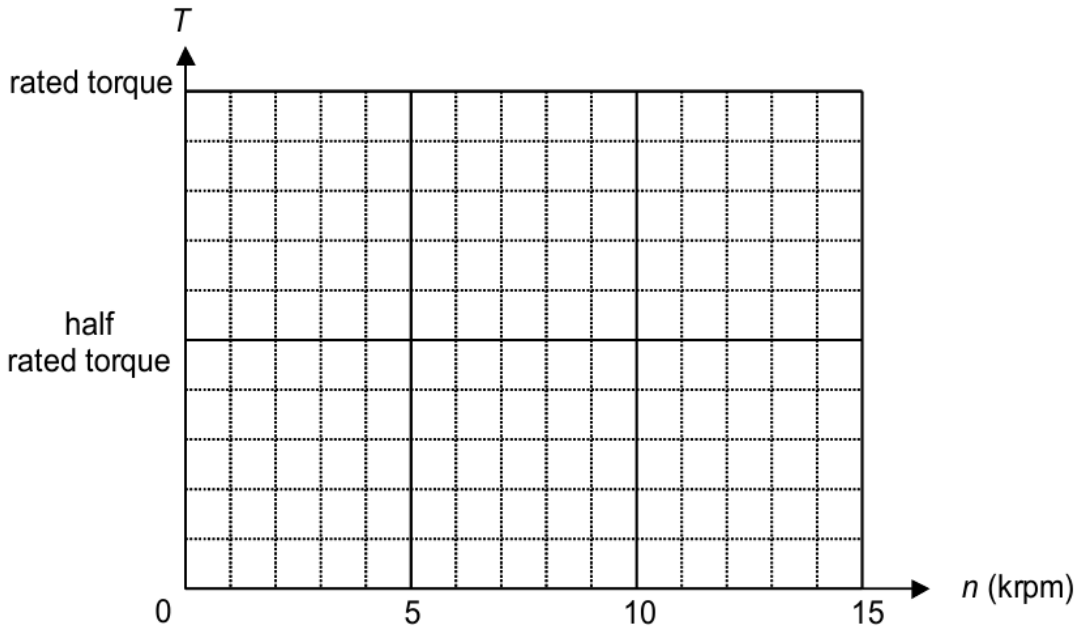
Jun24 Q3a figure (from the exam paper): three torque–speed axes (torque is labelled T on the paper) showing, left to right, a vertical line, a straight falling line, and the induction-motor curve

PQ 4.3 (practice questions). A 500W DC machine has a no-load speed of 12,000 rpm and a rated speed of 10,000 rpm when operated at rated voltage.

a. What is the rated output power of the DC machine?
b. Calculate the rated output torque.
c. Calculate the motor speed (in rpm) corresponding to half rated torque.
d. Calculate the motor output power with half rated torque. Why is this higher than half rated power (250W)?

PQ 4.4 (practice questions). Consider the DC machine described in the question above except this time it is operated with half rated voltage.

a. What is the new no-load speed?
b. What is the new full-load speed?
c. What is the speed corresponding to half rated output torque?
d. Using the template provided below: (i) Draw the torque versus speed curves corresponding to operation with rated and half rated voltage. (ii) Draw circles representing the full-load torque, half-load torque and no-load operating points for the both torque versus speed curves.



PQ 4.4 template (from the practice questions): torque axis marked rated torque and half rated torque (labelled T on the sheet), speed axis n from 0 to 15 krpm

PQ 4.7 (practice questions). A 0.75 kW induction machine is operating from a 110 V, 60 Hz AC power supply. Its synchronous speed is 1200 rpm and its rated speed is 1100 rpm.

a. Calculate the required supply voltage and frequency to give a synchronous speed of 900 rpm.
b. With the conditions in part (a), calculate the speed corresponding to: (i) rated torque (ii) half rated torque

PQ 4.11 (practice questions). You are required to select an AC machine to drive a mechanical load. The AC machine must operate from the 50 Hz AC supply. It is desirable that the machine have a synchronous speed as close as possible to 630 rpm as possible. Select an appropriate number of poles for the motor and calculate the exact synchronous speed.

PQ 4.13 (practice questions). A 11 kW, eight pole synchronous motor is operating from the 50 Hz mains.

a. What is its synchronous speed (in rpm)?
b. What is its rated speed (in rpm)?
c. What is its rated torque?
d. What is its speed at half rated torque?
e. What is its output power at half rated torque?

PQ 5.8 (practice questions). For the machine in the previous question, calculate

a. its no-load speed
b. its rated efficiency $= \omega/\omega_{NL} = n/n_{NL}$
c. the following when the speed is zero (stall condition): (i) induced voltage (ii) input current (iii) output torque (iv) input power (v) output power (vi) efficiency
d. the speed (in rpm) when the torque is 0.2 Nm

The previous question (PQ 5.7) reads: A DC machine has a back-emf constant of 0.24 V/(rad/s) and a resistance of 0.5 ohms. It is operating from a 12V DC power supply. At its rated speed of 400 rpm, calculate its (a) rated back-emf voltage (b) rated current (c) rated torque (d) rated output power $P_{OUT} = \tau\omega$ (e) rated output power $P_{OUT} = EI$ (f) rated input power (g) rated efficiency $= P_{OUT}/P_{IN}$ (h) rated efficiency $= E/V$.

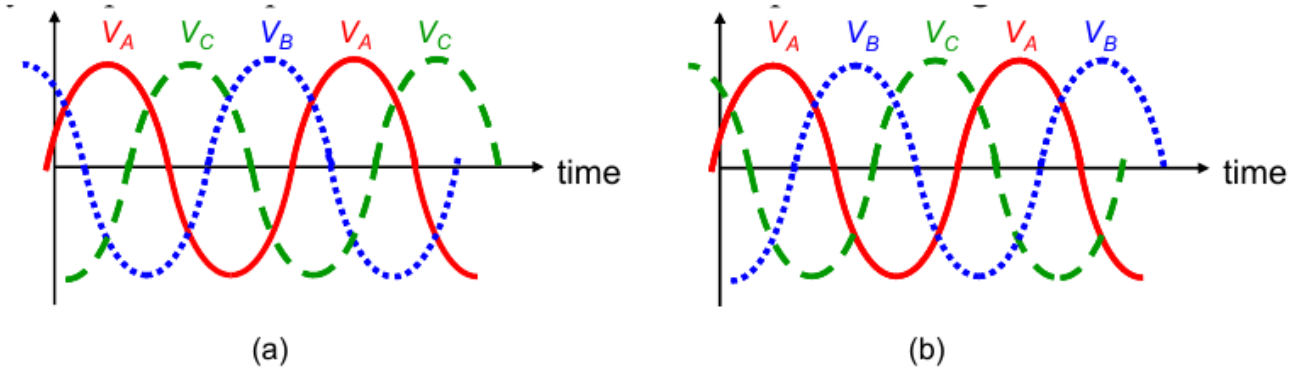
PQ 5.10 (practice questions). A DC machine is connected to a 24 V supply. When the machine is held at zero speed (stall), a current of 35 A flows and the machine produces a torque of 15 Nm. Find:

a. the no-load speed (in rpm)
b. the machine resistance
c. the machine output power at 400 rpm

PQ 5.19 (practice questions). A DC machine is driven by a small wind turbine. When the wind turbine is spinning at 150 rpm the measured open-circuit voltage is 10.5 V. The DC machine has an armature resistance of 0.4 ohms. The machine is then connected to a battery which acts like a constant voltage source of 14.6 V.

a. What is the back-emf constant of the machine?
b. At what turbine speed in rpm does the induced voltage just equal the battery voltage? What is the machine output current under this condition?
c. At what turbine speed in rpm, does the machine output current equals 10A?
d. For the condition in part (c), what is the efficiency of the machine?
e. In a practical application, a diode would be placed in series with the output of the wind generator. This diode allows power to flow from the generator into the battery but not vice-versa. Why is this diode required?

PQ 6.4 (practice questions). Identify the phase sequence for each of the three-phase voltage waveforms shown below:



PQ 6.4 figure (from the practice questions): two sets of three-phase voltage waveforms, (a) peaking in the order VA, VC, VB and (b) VA, VB, VC

PQ 6.5 (practice questions). You connect a three-phase AC induction machine to the mains supply. On turning it on you find that it rotates in the wrong direction. How do you reverse the direction of rotation of the machine?

PQ 6.7 (practice questions). Are the following a valid number of poles for an AC machine?

a. 2 poles (b) 15 poles (c) 4.5 poles (d) 16 poles

PQ 6.9 (practice questions). Synchronous generators are designed to feed power into the power system. In order to do so, they need to operate at the same frequency as the power grid.

a. In Australia, what is the highest speed a synchronous generator can operate at, and how many poles does this correspond to?
b. You are designing a power-station generator for a hydro-electric plant in New South Wales. It is designed to operate at 100 rpm. How many poles should this be designed

for?

PQ 5.11 (practice questions, extra). A DC machine has a no-load speed of 10,000 rpm and a rated speed of 8,000 rpm when operating with its rated supply voltage of 20 V. It has a resistance of 0.5 Ω.

a. Calculate the input current and output torque when the machine is at zero speed.
b. Calculate the input current and output torque when the machine is at rated speed.
c. Sketch the machine voltage-current and torque-speed characteristics on the template provided. Label the no-load, full-load and stall operating points on both graphs and indicate the corresponding induced voltage, current, torque and speed on the appropriate axes.
d. By observation from the graphs (without doing any calculations), what is the current and speed corresponding to operation at half rated torque?

PQ 5.11 template (from the practice questions): I against E axes and torque (labelled T on the sheet) against n axes

PQ 5.12 (practice questions, extra). Consider the machine in the previous question. On the axes provided sketch the torque versus speed characteristics when operating at rated supply voltage and half rated supply voltage. For both curves show the values of the stall current and torque, and the no-load induced voltage and speed.

PQ 5.12 template (from the practice questions): I against E axes and torque (labelled T on the sheet) against n axes

New questions

N1. [4 marks] A three-phase induction motor is rated at 1140 rpm on a 60 Hz supply.

a. Find the number of poles and the synchronous speed. [2]
b. Find the ideal (maximum) efficiency at rated speed. [1]
c. Assuming a linear τ–n curve, find the speed at 60 % of rated torque. [1]

Circuit for New question 2: PM DC motor with R = 0.25 Ω on 48 V, rated current 12 A

N2. [6 marks] A PM DC motor has a rated voltage of 48 V, an armature resistance of 0.25 Ω, a no-load speed of 3000 rpm at rated voltage, and a rated current of 12 A.

a. Find the back-emf constant k. [1]
b. Find the rated torque, rated speed (rpm) and rated efficiency. [3]
c. Find the stall current and stall torque, and express the stall torque as a multiple of rated torque. [1]
d. Find the maximum possible output power and the speed at which it occurs. Why is this not a sensible operating point? [1]

N3. [6 marks] (chains from N2) The motor in N2 drives a constant-torque load equal to its rated torque.

a. The supply is reduced to 24 V. Find the new no-load speed, the operating speed and the efficiency. [3]
b. What supply voltage gives an operating speed of 1000 rpm with this load? [2]
c. At 24 V, the load torque falls to half of rated. Find the speed. [1]

Circuit for New question 4: wind-driven DC machine (k = 0.5 V/(rad/s), R = 0.8 Ω) connected directly to a 24 V battery

N4. [6 marks] A DC machine with k = 0.5 V/(rad/s) and R = 0.8 Ω is driven by a small wind turbine and connected directly to a 24 V battery.

a. Find the turbine speed (rpm) at which the charging current just becomes zero. [1]
b. Find the speed needed to charge at 5 A. For this condition find the shaft torque, the mechanical input power, the electrical power into the battery and the efficiency. [3]
c. With no diode fitted, the wind drops and the turbine slows to 400 rpm. Find the current and its direction, and the torque. Explain what is happening. [2]

N5. [5 marks] A 4-pole, 415 V, 50 Hz induction motor has a rated speed of 1440 rpm. It is fed from a V/f inverter.

a. Find the inverter frequency and voltage so that the motor runs at 900 rpm at rated torque. [3]
b. With those settings, find the speed at half rated torque. [1]
c. A 6-pole synchronous motor with the same 415 V, 50 Hz rating must also run at 900 rpm. Find the frequency and voltage. [1]

N6. [4 marks] (words only)

a. A three-phase motor is connected with supply phases a, b, c to terminals A, B, C, and runs clockwise. An electrician reconnects it as b to A, c to B, a to C. Which way does it now run? Explain. [2]
b. A hydro generator for the Australian grid is designed to turn at 100 rpm. How many poles does it need? How many would it need for the same speed on a 60 Hz grid? [1]
c. In a 4-pole machine on 50 Hz, how many electrical cycles occur per mechanical revolution of the field? [1]

6. Quick recall check

1. What is the synchronous speed of an 8-pole machine on 60 Hz?
2. A DC motor has $n_{NL} = 3000$ rpm and $n_R = 2700$ rpm at 24 V. What are n_{NL} and n_R at 12 V?
3. At stall, what are E and I for a DC motor?
4. Why must V be reduced with f in an AC drive?
5. An induction motor runs at 1470 rpm on 50 Hz. What is its ideal maximum efficiency?